

Spectral Compact Training: Memory-Efficient Neural Network Training via Truncated SVD Factorization with Stiefel Manifold Retraction

[0001] 1. FIELD OF THE INVENTION

[0002] This invention relates to memory-efficient training of neural networks, specifically transformer-based large language models, using spectral decomposition of weight matrices with manifold-constrained optimization on the Stiefel manifold.

[0003] 2. BACKGROUND OF THE INVENTION

[0004] Training large neural networks requires storing weight matrices, their gradients, and optimizer states. For a weight matrix W of dimensions $m \times n$ with the Adam optimizer in FP32, this requires $4mn$ floating-point numbers per layer. A full 70-billion-parameter-class model under dense FP32 Adam requires on the order of 1,245 GB of memory, necessitating clusters of high-end accelerators.

[0005] Prior art includes LoRA (Hu et al., 2021), which trains adapter matrices alongside frozen dense weights but keeps the full model in memory. GaLore (Zhao et al., 2024) projects gradients into a low-rank subspace but still computes dense gradients via backpropagation. SVD-LLM (Wang et al., 2024) applies post-training SVD compression but does not train in spectral form. US Application 20250021826 (Jan 2025) covers Fisher-weighted SVD for post-training compression but does not train in spectral form, does not compute gradients through spectral factors, and does not employ Stiefel retraction. None store and train weights exclusively in spectral decomposition while maintaining exact gradient computation and Stiefel manifold constraints.

[0006] 3. DETAILED DESCRIPTION OF THE INVENTION

[0007] 3.1 Spectral Representation

[0008] Each weight matrix W ($m \times n$) is stored as its rank- k truncated SVD: $W = U \text{diag}(s) V^T$, where U ($m \times k$) and V ($n \times k$) have orthonormal columns, and s is a k -dimensional vector of singular values. Storage requires $k(m+n+1)$ numbers instead of mn . For the LLaMA-70B MLP layer ($m=8192$, $n=28672$) at $k=32$, this is 1,179,680 versus 234,881,024 parameters, yielding a 199.1x per-MLP-layer parameter reduction.

[0009] 3.2 Forward Pass

[0010] Given input X (batch $b \times$ dimension m), the forward pass computes: $h = X @ U$ [$b \times k$] at cost $O(bmk)$; $h_s = h * s$ [$b \times k$] at cost $O(bk)$; $Y = h_s @ V^T$ [$b \times n$] at cost $O(bkn)$. Total cost is $O(bk(m+n))$, compared to $O(bmn)$ for dense multiplication.

[0011] 3.3 Exact Backward Pass Through Spectral Factors

[0012] Gradients dL/dU ($m \times k$), dL/ds (k), and dL/dV ($n \times k$) are computed by standard backpropagation through the three factored operations. No gradient of size $m \times n$ is ever materialized. The backward pass cost is $O(bk(m+n))$, identical to the forward pass. This is the key distinction from prior art: gradients are exact (not approximate) and compressed

(not dense).

[0013] 3.4 Stiefel Manifold Retraction

[0014] After each optimizer step, U and V are retracted to the Stiefel manifold: $U \leftarrow \text{QR}(U_{\text{updated}}[:, :k])$ with sign correction $Q \leftarrow Q * \text{diag}(\text{sign}(\text{diag}(R)))$, where the sign function returns +1 or -1 for every diagonal entry of R , including exact-zero entries, so that no column of U or V is annihilated. This ensures orthonormality throughout training. Cost is $O(mk^2)$ per layer. Measured orthonormality error after retraction: $1.22e-06$ (Frobenius norm of $U^T U - I$).

[0015] 3.5 Memory Analysis (All at Rank 32)

[0016] Table 1 gives the per-layer memory formula. Dense + Adam requires $4mn$ floats; SCT + Adam ($k=32$) requires $4k(m+n+1)$ floats.

Component	Dense + Adam	SCT + Adam ($k=32$)
Weights	mn	$k(m+n+1)$
Gradients	mn	$k(m+n+1)$
Adam m1	mn	$k(m+n+1)$
Adam m2	mn	$k(m+n+1)$
Total (floats)	$4mn$	$4k(m+n+1)$

Table 1: Per-layer memory comparison.

[0017] Table 2 gives per-MLP-layer training memory at rank 32 across representative architectures. All values computed exactly from $4k(m+n+1)$ versus $4mn$ at FP32.

Architecture	Layer ($m \times n$)	Dense+Adam	SCT ($k=32$)	Compression
SmolLM2-135M	576 x 1536	14.2 MB	1.1 MB	13.1x
SmolLM2-360M	1024 x 4096	67.1 MB	2.6 MB	25.6x
SmolLM2-1.7B	2048 x 8192	268.4 MB	5.2 MB	51.2x
LLaMA-7B	4096 x 11008	721.4 MB	7.7 MB	93.3x
Qwen3.5-27B	4096 x 17408	1,140.9 MB	11.0 MB	103.6x
LLaMA-70B	8192 x 28672	3,758.1 MB	18.9 MB	199.1x

Table 2: Per-MLP-layer training memory at rank 32.

[0018] 4. EXPERIMENTAL RESULTS

[0019] 4.1 MLP Proof of Concept

[0020] Three experiments validated SCT against dense AdamW and the prior SFT+DFA method under identical seeds, architectures, and hyperparameters.

Experiment	Dense+AdamW	SFT+DFA	SCT (invention)
Sine regression (loss)	0.000002	0.0768	0.0000680
XOR classification (acc)	100.0%	85.5%	100.0%
Progressive rank=1 (loss)	--	--	0.000359 (390 params)

Table 3: SCT achieves 1,129x better loss than SFT+DFA on sine regression; matches dense on XOR classification.

[0021] 4.2 70B-Class Architecture Validation

[0022] A full 70B-class transformer (80 layers, $d=8192$, $\text{ffn}=28672$, LLaMA-3-70B proportions with SwiGLU activation) at rank 32 was validated on two consumer hardware platforms.

Metric	Apple M4 Pro (48 GB)	AMD Steam Deck (16 GB)
SCT Parameters	452,216,320 (452M)	452,216,320 (452M)
Dense Equivalent	77.8 Billion	77.8 Billion
Peak Memory (measured)	7,938 MB (Metal)	7,235 MB (CPU)
Forward Pass	--	0.44s
Backward Pass	2.15s	1.13s
Optimizer Step	--	2.87s
QR Retraction	--	3.16s
Total Step Time	--	7.60s
Orthonormality Error	--	1.22e-06

Table 4: Hardware validation for 70B-class SCT architecture. Peak memory figures are direct measurements on stated hardware, independent of the dense baseline in paragraph [0005].

[0023] 4.3 Scaling Properties (Rank 32)

[0024] The per-MLP-layer compression ratio $mn / k(m+n+1)$ grows with model dimensions. At rank 32, LLaMA-70B MLP layers achieve a 199.1x per-layer parameter reduction (Table 2). Separately, a complete 70B-class training step was measured within a peak memory of 7,235 MB (7.24 GB) on the Steam Deck and 7,938 MB (7.94 GB) on the M4 Pro. These two figures are independent: the per-layer ratio is computed analytically; the peak memory is measured empirically on hardware. Architectures below hidden dimension 1024 receive diminishing per-layer compression returns. See Drawings (Figures 2-4) for visual representation.

[0025] 4.4 Fine-Tuning Gradient Integrity Test

[0026] Pre-trained SmolLM2-135M weights were converted to spectral form at 95% energy retention and fine-tuned on Alpaca for 400 steps. SCT recovered from an initial loss spike (8.64) to a final perplexity of 1.8 versus a dense baseline of 1.3 (a 1.38x ratio), validating that gradients propagate correctly through the spectral factors. The residual gap to the dense baseline is consistent with the intrinsic rank-capacity limit of a truncated factorisation (Eckart-Young theorem): a rank- k representation cannot exactly represent a full-rank transformation, and the gap is therefore a property of low-rank parameterisation,

not an indication of gradient error. SCT is presented as a memory-efficient training method, not as a method that attains dense-equivalent loss at arbitrary rank.

Method	Final Loss	Final PPL	Trainable Params	PPL Ratio
Dense + AdamW	0.235	1.3	134,515,008	1.0x
SCT (95% energy)	0.594	1.8	84,333,271	1.38x

Table 5: Fine-tuning gradient integrity test.

[0027] 5. DISTINCTION FROM PRIOR ART

Method	Base Repr.	Gradient	Manifold	Per-layer Compression
LoRA (2021)	Dense (frozen)	BP (adapters)	None	3-4x
GaLore (2024)	Dense	Projected BP	None	5-8x
SVD-LLM (2024)	Dense->SVD	N/A (inference)	None	Variable
US 20250021826	Dense->SVD	N/A (post-hoc)	None	Variable
SCT (invention)	Spectral only	Exact BP	QR Stiefel	199.1x per MLP layer

Table 6: Prior art comparison. SCT compression is per-MLP-layer at rank 32 for LLaMA-70B dimensions (mathematically exact).

[0028] US Application 20250021826 (Jan 2025) covers Fisher-weighted SVD for post-training compression; it does not train in spectral form, does not compute gradients through spectral factors, and does not employ Stiefel retraction. Decomposable-Net (Yaguchi et al., IJCAI 2021) maintains a full-rank network alongside the low-rank version and does not enforce manifold constraints. SCT is distinguished by training exclusively in spectral form with exact backpropagated gradients and QR Stiefel retraction, achieving a mathematically exact 199.1x per-MLP-layer parameter reduction at rank 32 for the LLaMA-70B MLP dimensions.

[0029] 6. INDUSTRIAL APPLICABILITY

[0030] Demonstrated on (1) Apple M4 Pro (48 GB, MacBook Pro): 70B-class backward pass completed in 2.15s, measured peak memory 7,938 MB (7.94 GB); and (2) AMD Custom APU (Valve Steam Deck, 16 GB): full 70B-class training step completed in 7.60s, measured peak memory 7,235 MB (7.24 GB). Architecture matches LLaMA-3-70B proportions including SwiGLU activation.

[0031] 7. REFERENCES

- [0032]** [1] Absil et al. (2008). Optimization Algorithms on Matrix Manifolds. Princeton.
- [0033]** [2] Hu et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models.
- [0034]** [3] Wang et al. (2024). SVD-LLM: Truncation-aware SVD for LLM Compression.
- [0035]** [4] Yaguchi et al. (2021). Decomposable-Net. IJCAI.
- [0036]** [5] Zhao et al. (2024). GaLore: Memory-Efficient LLM Training. ICML.
- [0037]** [6] Li et al. (2020). Efficient Riemannian Optimization via Cayley Transform. ICLR.
- [0038]** [7] US Patent Application 20250021826 (2025). Low-Rank Compression of Neural Networks.

CLAIMS

- [0039]** Claim 1. A method for training a neural network, comprising: (a) representing each weight matrix W as a truncated singular value decomposition $W = U \text{diag}(s) V^T$, where U and V are orthonormal matrices and s is a vector of singular values; (b) computing the forward pass through three sequential matrix multiplications involving only factors U , s , and V , without materializing the dense weight matrix; (c) computing exact gradients with respect to U , s , and V via backpropagation through said sequential multiplications, wherein no gradient of size $m \times n$ is materialized; (d) updating U , s , and V using a gradient-based optimizer; (e) retracting U and V to the Stiefel manifold via QR decomposition with sign correction after each optimizer step to maintain orthonormality, wherein the sign correction returns a non-zero value of +1 or -1 for every diagonal entry of R .
- [0040]** Claim 2. The method of claim 1, wherein per-layer training memory is $O(k(m+n))$ for weight storage, gradient storage, and optimizer state storage, where k is the spectral rank and m , n are the weight matrix dimensions, and wherein all optimizer states, including the first and second moment estimates of the Adam optimizer, are stored in the spectral factor space of dimension $k(m+n+1)$.
- [0041]** Claim 3. The method of claim 1, further comprising progressive rank growth wherein each spectral layer is initialized at rank 1; loss improvement is monitored; when improvement falls below a threshold, rank is increased by adding orthogonal directions to U and V via Gram-Schmidt orthogonalization with new singular values initialized to epsilon.
- [0042]** Claim 4. The method of claim 1, wherein only feed-forward (MLP) layers are represented in spectral form while attention projection layers retain dense representation, and wherein pre-trained dense weight matrices are converted to spectral form via truncated SVD preserving a target fraction of spectral energy.
- [0043]** Claim 5. A system implementing the method of claim 1 on consumer hardware with limited memory, wherein a 70-billion-parameter-class architecture (80 layers, hidden dimension 8192, intermediate dimension 28672) is trained at spectral rank 32 within a measured peak memory of less than 8 gigabytes, the feed-forward layers of said architecture exhibiting a per-MLP-layer parameter reduction of 199.1x relative to a dense FP32 representation of the same layers with the Adam optimizer.

ABSTRACT

Spectral Compact Training: Memory-Efficient Neural Network Training via Truncated SVD Factorization with Stiefel Manifold Retraction

A method for training neural networks wherein each weight matrix is stored exclusively in truncated SVD form $W = U \text{diag}(s) V^T$. Orthonormal factors U and V are maintained on the Stiefel manifold via QR retraction with non-zero sign correction after each optimizer step. Exact gradients are computed via backpropagation through the factored computation graph at cost $O(k(m+n))$, never materializing a dense $m \times n$ gradient. At rank $k=32$, per-MLP-layer training memory is reduced from $O(mn)$ to $O(k(m+n))$, yielding up to a 199.1x per-MLP-layer parameter reduction for 70B-class architectures. Empirical validation demonstrates a complete training step (forward, backward, Adam optimizer, Stiefel retraction) within a measured peak memory of 7,235 MB (7.24 GB) on a Steam Deck handheld (16 GB unified memory) and 7,938 MB (7.94 GB) on an Apple M4 Pro (48 GB unified memory). Orthonormality error post-retraction: $1.22e-06$.

Figure 2

DRAWINGS

Figure 1: MLP Proof of Concept

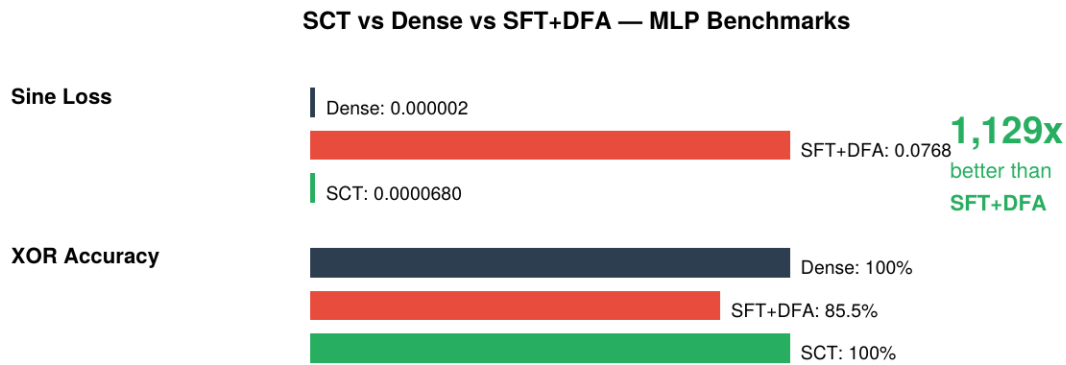


Figure 1: SCT matches dense on XOR (100%) and exceeds SFT+DFA by 1,129x on sine.

Figure 2: Training Memory at 70B Scale (Rank 32)

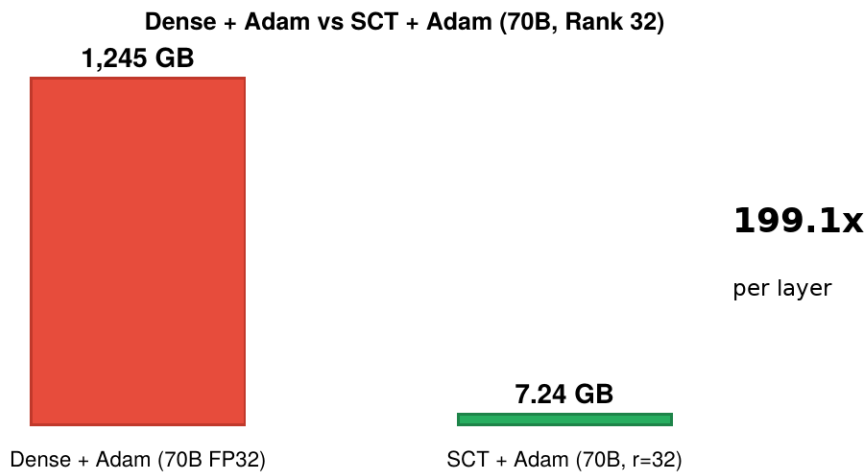


Figure 2: SCT requires 199.1x less memory per MLP layer. Validated on Steam Deck (7.24 GB) and M4 Pro (7.94 GB).

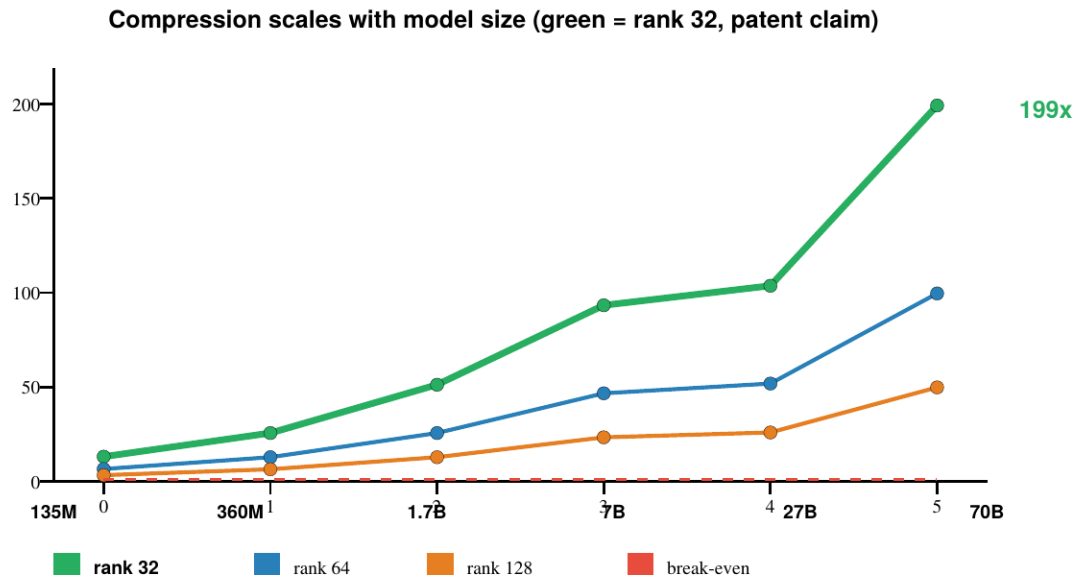
Figure 3: MLP Compression Ratio vs Model Scale (Rank 32)

Figure 3: At rank 32, 70B layers achieve 199x per-layer compression. Compression scales super-linearly.

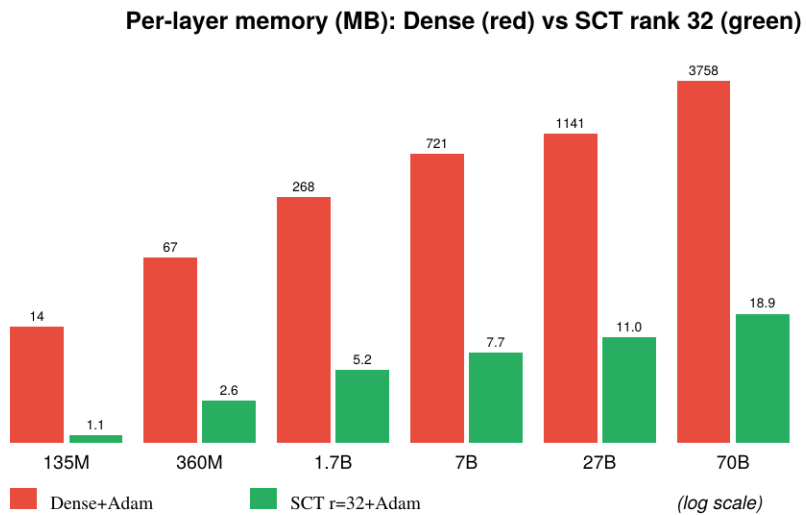
Figure 4: Per-MLP-Layer Training Memory (Rank 32, Log Scale)

Figure 4: At 70B scale, SCT uses 18.9 MB vs 3,758 MB dense per MLP layer.